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THE EFFECT OF NEUTRON IRRADIATION ON
CERTAIN PROPERTIES OF REFRACTORY CONCRETES

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S u m m a r y

Unlike that of the materials used in the construction of nuclear reactors, the radiation resistance of concretes used for biological protection has been insufficiently studied. At present there is a tendency to use for biological protection mainly refractory concretes instead of the relatively expensive materials which are in short supply (steel, pig-iron, graphite, boron, etc.). For this reason, the present article sets out the results of investigations into the effects of reactor irradiation by neutrons on certain properties of refractory chromite concretes based on Portland cement and water-glass. The overall neutron flux was $2-2.4 \times 10^{21}$ neut/cm², and the irradiation temperature was up to 550°C.

It has been established experimentally that these concretes retain fairly high characteristics of strength and elasticity. The thermal conductivity and coefficient of thermal expansion vary only slightly. It is concluded that these concretes may be recommended for use in the biological shielding of nuclear reactors.

* * *

The information found in literature on the radiation resistance of concrete is rather contradictory. Furthermore it relates to comparatively low overall fluxes of low-energy neutrons (1,2). Hitherto, there had been no need to explore more deeply the radiation resistance of concrete: in thermal reactors, the leakage of neutrons into the shielding is not great; between the reactor vessel and the concrete of the biological shielding a thermal screen was installed to lower the radiation flux on the concrete to 10^{11} MeV/cm²/s. The latter figure, which has been recommended by a number of authors (3) from considerations of the favourable operating temperature in the shielding material (with a temperature drop of 30-60°C) was known to limit the overall neutron flux on the

shielding to values of between 10^{18} and 10^{19} neut/cm² (the operational life of the reactor was assumed to be between 10 and 20 years).

At the present time, in view of the increased construction of atomic power stations and the development of fast reactors (with their increased radiation effects on the shielding) there has been a tendency to replace the thermal shielding, made of costly materials which are difficult to obtain, by heat-resistant reinforced concrete (4). The best material for this purpose is heat-resistant reinforced concrete containing a chromite filler, which gives effective protection even when it contains no water (4). Constructions made of such concrete are used at temperatures of up to 1000 and 1100°C and with temperature drops of up to 600 and 700°C (5). The permissible radiation flux on the concrete, according to the operating temperature, is up to 10^{13} neut/cm²/s (4). In this way, to solve the problem of the practical utilisation of chromite concrete in shielding it is necessary to have information about the radiation resistance at overall fluxes of 10^{21} - 10^{22} neut/cm². (It is stated in the conclusion of (4) that the absence of information concerning the radiation resistance of concrete is the sole obstacle preventing its use for providing protection against intense radiation fluxes.)

The present article gives the results of investigations into the effect of radiation in a reactor on certain characteristics of specimens of chromite refractory concrete based on Portland cement and water-glass*. The chemical compositions of the materials investigated are given in Table 1.

The experimental test-pieces, in the form of briquettes 15 mm high and 15 mm in diameter were prepared as follows. The mixture,

* Some data on this question have already been published at the beginning of 1966 (6).

prepared in the form of a solution, was placed in separable metal moulds and packed hard by vibration for 30 seconds. Next, the specimens of chromite concrete with a Portland cement base were kept for 7 days in a humid atmosphere, whereas the concrete specimens based on water-glass were kept for 3 days in normal conditions, after which all of the specimens were dried out for 32 hours at a temperature of 100 - 110°C. After this the specimens were kept at a temperature of 20 - 25°C and a humidity of 60-70% until the commencement of the experiments.

Before the experiment was carried out, the specimens were broken up into three groups. The first group (subjected to radiation) was irradiated in special hermetic containers made of stainless steel, in the active zone of the reactor, at a temperature of 200 - 550°C.

A typical curve of the variation in the temperature is shown in Figure 1. Throughout the period of irradiation the specimens were exposed to an overall neutron flux of $2 - 2.4 \times 10^{21}$ neut/cm² at an average neutron energy of 0.23 MeV (including 30% neutrons with an energy higher than 0.7 MeV). The second group of specimens (the hot group), was also kept in hermetically welded containers (in order to eliminate the temperature effect), and the temperature and humidity conditions were similar to those of the irradiated group, i.e. the temperatures and the duration of their variations corresponded accurately to the temperatures and thermal variations of the irradiated group. The specimens of the third group (the "cold" group) were treated as reference specimens and were kept in normal conditions of temperature and humidity.

The removal of the test-pieces from the shells and the examination of their properties were carried out in heated chambers. The specimens were observed through a window of the chamber by means of binoculars (10 x magnification). No substantial damage or distortion was observed. It was noticed that one of the end surfaces in practically all of the test-pieces subjected to

radiation or used as hot control specimens was more or less round (Figure 2). This may be explained by insufficient packing down of the mixture when the specimens were prepared (porous projections) and by the severe conditions in which they were maintained: the high temperatures and temperature variations.

The length and diameter of the specimens was measured with a micrometer and a slide caliper (with an accuracy of 0.05 mm). The average reduction in the linear dimensions (diameter and height) of the hot and irradiated specimens was 0.1 and 0.4%^{*} respectively compared with the cold specimens.

The specimens were weighed on an analytical balance accurate to within 0.05 g; this did not show any noticeable variation in the weight of the irradiated specimens in comparison with the "hot" reference specimens.

The thermal conductivity coefficient of the irradiated and reference specimens was determined with the same equipment in a hermetic box by using the constant heat flow method. The temperature of the end surfaces of the specimens was, during measurement, maintained at 20 - 30°C on one side and 120 - 130°C on the other, i.e. the average temperature was 70 - 80°C. A number of measurements were carried out on each specimen; tests were made on three specimens on each group.

The results of measurements are grouped together in Table 2, which shows that the coefficients obtained for thermal conductivity are close to the values given in (7); the slightly greater thermal conductivity is quite normal for the cold group and is explained by the greater humidity of the specimens in this group; within the limits of measurement accuracy (5 - 10%) the thermal conductivity of the chromite concrete does not vary after irradiation by a flux of $2 - 2.4 \times 10^{21}$ neut/cm².

* According to information from the Scientific Research Institute of Reinforced Concrete, the dimensions of the chromite concrete specimens based on Portland cement, which were reduced as a result of contraction, should after heating to 500°C be reduced by 0.2%.

The coefficient of thermal expansion was determined for the chromite concrete with a Portland cement base. Experiments were made on a dilatometer, and the elongations of the specimen were determined by indicators having a 0.002 mm graduation; the temperature was measured with chromel-alumel thermocouples. The highest temperature to which a specimen was heated was 500°C, the heating rate 6 - 7 deg./min. The specimens were held at temperatures of 100, 200, 300, 400 and 500°C for 15 min., after which the elongation was determined. After cooling the specimen was left for 12 hours in normal conditions of temperature and humidity, after which the second heating was carried out (using the first heating rate), during which measurements were also made.

Figure 3 shows graphs for the elongation of the specimens depending on the temperature. The values of the coefficient of linear thermal expansion calculated on the basis of these graphs are little different from those given in (7). The smaller elongations for the cold specimens during their first heat-up (see Figure 3a) are explained by the fact that some shrinkage takes place. No distinct differences were noted between the elongations of the irradiated and hot groups.

Strength tests were carried out on the irradiated and reference specimens using a distance-type tensile-testing machine at a temperature of 20°C. The rate of compression was 11 mm/min. During the test the compression graph was recorded. The results of the experiments are given in Table 3.

The Table shows that the strength of the irradiated specimens of chromite concrete based on Portland cement fell by 60% in relation to the material maintained in normal conditions, and by 39% in comparison with the concrete which was maintained at the temperature of irradiation of 200 - 550°C. The reduction in the strength of the specimens as a result of the influence of only the temperature factor is 34% (7). Consequently roughly 26% of the reduction in strength can be explained by the effect of

irradiation.

For the specimens of the chromite concrete based on water-glass, the strength after irradiation fell by 68 and 40% in comparison with the cold and hot specimens respectively. The reduction in strength as a result of heating was 47%, i.e. the reduction in strength on account of irradiation is 21% (see Table 3). The reduction in strength of the concrete after irradiation at 200 - 500°C which is about 60% should not give cause for concern. Calculations on shielding structures for temperature drops of up to 800°C (4,7) have shown that the maximum compressive forces obtained in the concrete are of the order of 20 - 30 kg/cm², i.e. with the minimum grade of chromite concrete 250 allowed by the standards (the calculated resistance to compression is 105 kg/cm²) ~70% of its strength is not used.

The variation in the strengths of the concrete when irradiated in conditions of high temperatures is best computed by using a factor. For chromite concrete based on Portland cement, irradiated with a flux of 2.4×10^{21} neut/cm² at a temperature of 200 - 550°C, this factor is 0.4. When it is taken into account the calculated strength R_{cm}^p ¹⁾ of grade 250 concrete will be $105 \times 0.4 = 42$ kg/cm², which is slightly higher than the indicated values of 20 - 30 kg/cm².

In view of the small dimensions of the specimens, it is not possible to use a compression diagram for calculating the modulus of elasticity. The change in this characteristic after irradiation can be assessed only indirectly by the variation in the angle of the slope of the compression curve. Table 3 shows that the average value of the tangent of the angle of slope of the compression curve for concrete based on Portland cement is barely different in the case of the irradiated and hot groups (0.665 and 0.686), but falls by about 20% in comparison with the cold group. For concrete based on water-glass, the tangent of the angle falls by 13% for the hot group and 34% for the irradiated group in comparison with the cold one.

¹⁾ compression

The data given in this article show that the specimens of chromite concrete, when irradiated with a flux of $2 - 2.4 \times 10^{21}$ neut/cm² at a temperature of 200 - 550°C, retain fairly high strength and elasticity characteristics. Furthermore, such characteristics of the concrete as thermal conductivity and the coefficient of thermal expansion do not vary to a marked degree. Consequently, the radiation resistance of this type of concrete does not restrict its utilization for the construction of thermal shielding around the nuclear reactor.

Work is at present continuing to explore the effect of irradiation on the shrinkage of concretes and on their strength and elastic/plastic properties.

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T a b l e s

Table 1

Chemical Composition of the Specimens

Concrete	Element, Weight %									
	H	O	Mg	Al	Si	Ca	Cr	Fe	Na	Others
Chromite concrete, Portland cement	0,70	37	11,54	5,14	2,64	4,63	30,7	6,45	-	1,20
Chromite concrete, water-glass	-	32,94	10,12	4,48	6,79	-	27,91	6,79	8,5	1,6

Table 2

Coefficient of thermal Conductivity of Chromite Concretes,
kCal/m/h/C°

Specimen Group	Portland cement concrete		Water-glass concrete	
	Average result for the specimen	Average result for the group	Average result for the specimen	Average result for the group
Irradiated	0,880 0,910 0,970	0,920	0,732 0,775	0,754
Hot	0,895 0,946 0,934	0,922	0,787 0,812 0,770	0,790
Cold	0,994 0,880 0,970	0,945	0,990 0,942 1,02	0,978

Table 3

Strength and Elastic Properties of Concretes

Chromite concrete specimens with Portland cement			Chromite concrete specimens with water-glass		
Irradiated	Hot	Cold	Irradiated	Hot	Cold
$\frac{56,6^*}{0,64}$	$\frac{110,4}{0,77}$	$\frac{124,6}{0,187}$	$\frac{82,1}{0,615}$	$\frac{90,6}{0,682}$	$\frac{209,5}{-}$
$\frac{62,3}{0,68}$	$\frac{95,1}{0,725}$	$\frac{126,3}{0,885}$	$\frac{71,3}{0,545}$	$\frac{134,8}{-}$	$\frac{262,7}{0,885}$
$\frac{41,3}{0,68}$	$\frac{81,5}{0,725}$	$\frac{94}{0,68}$	$\frac{74,7}{0,58}$	$\frac{141,6}{0,853}$	$\frac{286,5}{0,885}$
$\frac{53,8}{0,68}$	$\frac{78,1}{0,634}$	$\frac{126,3}{0,885}$	-	$\frac{141,6}{0,767}$	$\frac{207,2}{0,885}$
$\frac{37,4}{0,612}$	$\frac{35,1}{-}$	$\frac{126,3}{0,817}$	-	-	-
$\frac{37,4}{0,68}$	$\frac{101,9}{-}$	$\frac{78,1}{-}$	-	-	-
-	$\frac{51}{0,578}$	$\frac{163,1}{-}$	-	-	-
Average Value					
$\frac{48,1}{0,665}$	$\frac{79}{0,686}$	$\frac{119,8}{0,816}$	$\frac{76,1}{0,58}$	$\frac{127,9}{0,767}$	$\frac{241,5}{0,885}$
Relative Value, %					
$\frac{40}{82}$	$\frac{66}{84}$	$\frac{100}{100}$	$\frac{32}{66}$	$\frac{53}{87}$	$\frac{100}{100}$
* The numerator of the section is the strength of the specimen in kg/cm ² ; the denominator is the tangent of the angle of the slope of the compression curve on the compression diagram.					

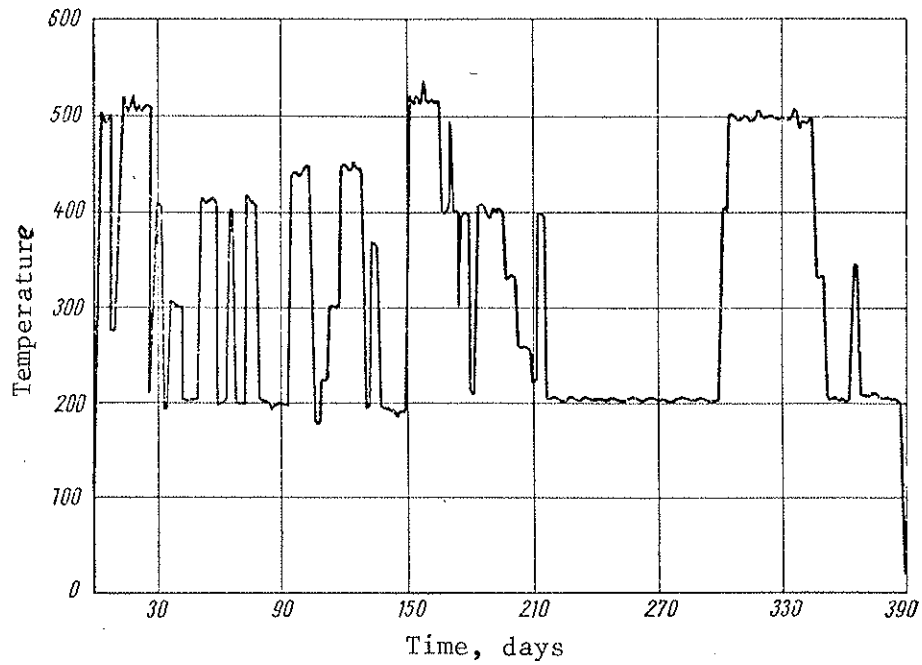


Figure 1: Curve of the variation in temperature of the specimens during irradiation.



Figure 2: Specimens of chromite concrete based on Portland cement after removal from the container.

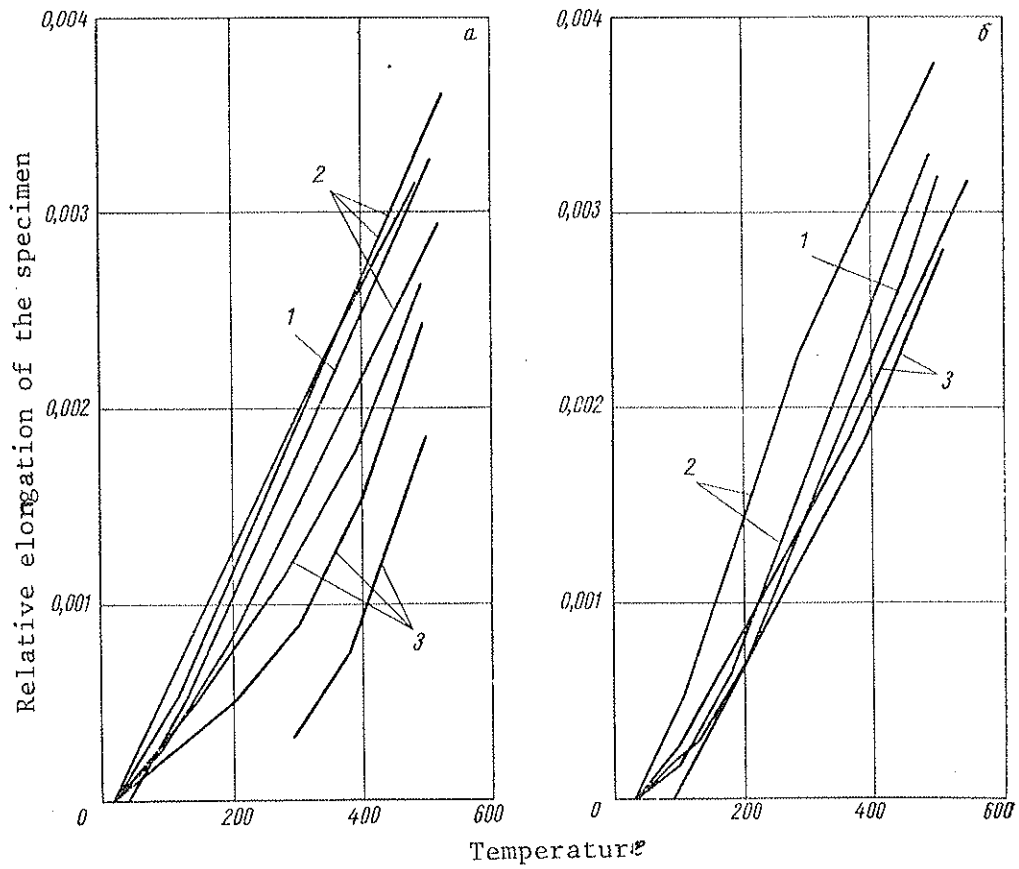


Figure 3: Dependence of the relative elongation of the specimens on the temperature:

- a - initial heating
- b - secondary heating
- 1 - irradiated specimens
- 2 - hot specimens
- 3 - cold specimens